

Lab 1 – PatientPath Descriptive Paper

Mia Lai

Old Dominion University

CS411

Dr Sumaya Sanober

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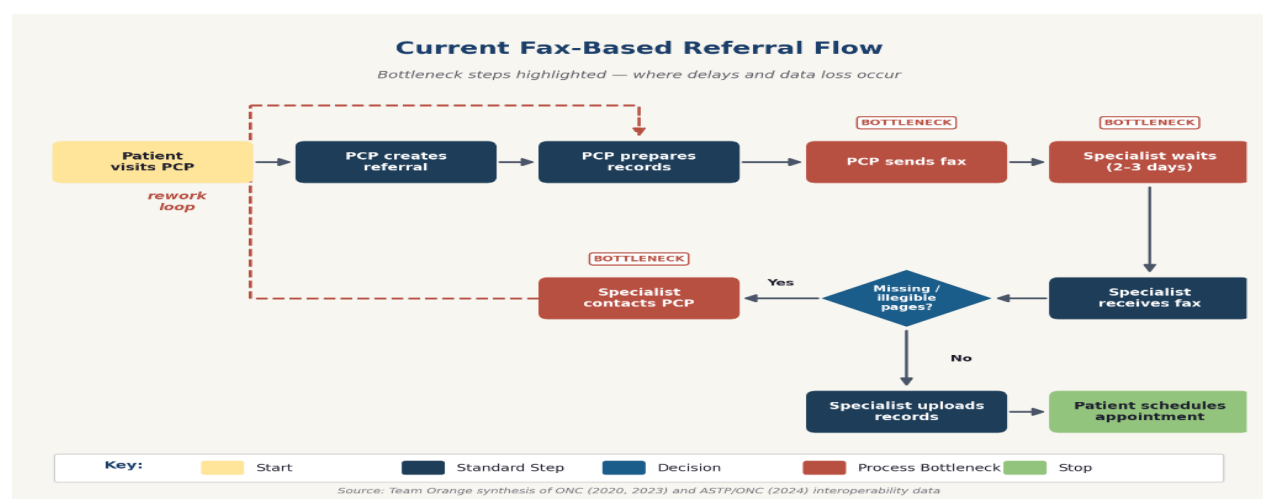
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1 Introduction

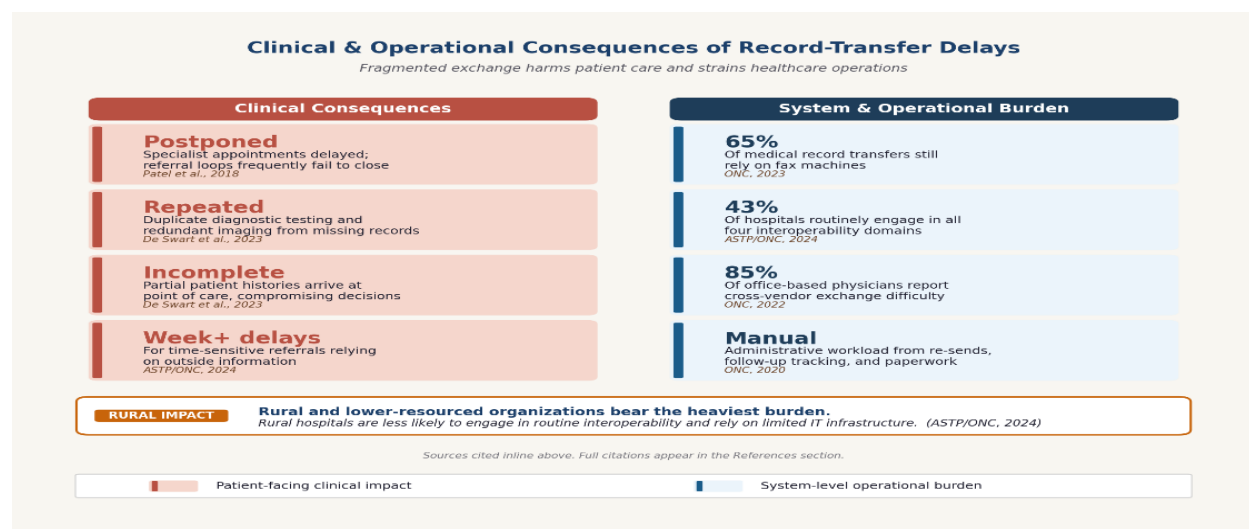
In the United States, healthcare continues to rely on outdated mechanisms for exchanging patient information. This creates measurable delays, administrative burden, and clinical risks. Despite the widespread availability of digital alternatives, approximately 65% of medical record transfers still occur by fax (ONC, 2023). Only 43% of hospitals routinely utilize all four core interoperability domains. These domains are sending, receiving, and integrating records (ASTP/ONC, 2024). The gap widens especially at the edges of the network. Only 16% of hospitals regularly share records with long term or post acute providers that are outside their own network. 17% share records with behavioral health providers (ASTP/ONC, 2024). Among office-based physicians, 85% report difficulty transferring records across different EHR vendors. 71% work with referral partners who lack exchange capability at all (ONC, 2022). This results in a fragmented national system in which critical patient information is too often locked behind incompatible software, proprietary formats, and paper type workflows (see Figure 1).

Figure 1
Fax Flow



These gaps translate into real harm for healthcare providers, their support staff, and their patients. Time sensitive referrals stall while staff resend, have to send multiple requests, and manually track status of paperwork (ONC, 2020). Handoffs between the primary care and specialists are incomplete, and referral loops frequently fail to close (Patel et al., 2018). Missing information leads to unnecessary repeat diagnostics, duplicate imaging, and redundant orders for various lab work ups (De Swart et al., 2023). This drives up overall costs while delaying care. The burden falls hardest on rural and lower resourced organizations, which are the least likely to engage in routine interoperability (ASTP/ONC, 2024) and the most likely to be forced to operate on a restricted IT infrastructure and erratic connectivity. A functional solution must therefore do more than just digitize existing workflows. It must be secure, interoperable with existing EHR systems, usable in real world network environments, and align with HIPAA requirements at every layer

Figure 2
Impact



A meaningful solution in this space must address six interlocking problem characteristics. Current transfer processes are inefficient, often delaying or corrupting records as they move between providers, and the systems involved are largely disconnected, with limited interoperability across organizations. Many still rely on outdated security protocols that leave patient privacy exposed, while poor usability slows adoption among clinicians who have little time to learn cumbersome tools. Rigid architectures also make it difficult for these systems to scale alongside growing healthcare networks, and connectivity constraints rule out online-only tools for rural deployment. A platform that addresses only some of these characteristics, such as one that achieves interoperability but fails in low-bandwidth environments, still leaves underserved the very populations who need it most and stand to benefit the most.

PatientPath is our proposed solution. It provides a secure, centralized, web based record sharing and referral coordination platform designed to bridge organizations and the existing EHR systems rather than replace them. Being built on an API first architecture with FHIR R4 and HL7 compatibility, PatientPath interweaves real time digital record exchange, automated referral tracking, role based access control, end to end encryption, and offline synchronization for rural and low bandwidth clinics. This is specifically designed to target hospitals, private practices, specialists, rural providers, and patients as well. PatientPath is engineered to be HIPAA aligned by design. The remainder of this document details PatientPath's features, architecture, case study, and describes a functional prototype built to demonstrate the platform's core characteristics in a representative rural referral scenario.

2 Product Description

PatientPath is a secure, web-based platform that moves medical record exchange off the fax machine and onto real-time digital sharing, with referral coordination handled automatically. It isn't meant to replace the EHR systems providers already rely on. Instead, it works between organizations as an interoperability layer, pulling records from existing systems, converting them to FHIR R4, and sending them across organizational lines through one API gateway. Providers, administrative staff, and patients each use an interface designed for their role, and every transaction is encrypted, logged, and tied to the patient's explicit consent.

PatientPath aims to remove the delays, lost documents, and administrative overhead that come with fax-based workflows, while also closing the privacy and security gaps those older channels create. Its objectives are to (1) enable real-time record exchange between organizations using FHIR R4 and HL7 v2.x, without asking providers to give up their current EHRs; (2) complete the referral loop with a single-form workflow that tracks status automatically, flags stalled referrals, and escalates them to supervisors; (3) meet HIPAA-aligned security standards through end-to-end encryption, role-based access control, multi-factor authentication, and a tamper-proof audit trail; (4) keep the platform usable in rural or low-bandwidth settings by caching data locally and syncing queued changes once a connection returns; and (5) maintain 99.9% uptime on cloud-hosted infrastructure that scales horizontally.

2.1 Key Product Features and Capabilities

PatientPath delivers eight integrated capabilities that together replace the fax workflow and address the six problem characteristics identified in Section 1. The user-facing features include secure record upload, real-time record sharing across organizations, automated referral tracking, patient consent management, transfer status visibility, multi-factor authentication, and offline sync resilience. The infrastructure features include FHIR/HL7 integration, an EHR adapter layer, end-to-end encryption, role-based access control, audit logging, and a high-availability cloud architecture.

- **Secure Record Upload** — Files are validated for format and completeness, encrypted with AES-256 via cloud KMS, and stored in encrypted document storage; uploads complete within five seconds.
- **Real-Time Record Sharing** — A scoped, time-limited IAM access token is issued to the receiving organization on referral creation; both provider and specialist dashboards update instantly with no manual handoff.
- **Automated Referral Tracking** — Single-form referral initiation is followed by automatic status transitions (Pending → In Review → Accepted → Complete); referrals inactive for 48 hours generate admin alerts, with supervisor escalation at 72 hours.
- **Patient Consent Management** — Patients grant and revoke record access from the patient portal; revocations propagate to all active IAM sessions in real time, immediately revoking the associated access tokens.

- **Multi-Factor Authentication** — TOTP two-step verification is required at every login; three failed attempts trigger account lockout and an admin alert within 30 seconds.
- **Offline Sync Capability** — Recently accessed records are cached locally during outages; queued writes reconcile with the server within 60 seconds of reconnection, with the server state acting as the source of truth in conflict resolution.
- **FHIR / HL7 Integration** — A configurable translation layer maps EHR data to FHIR R4 and supports HL7 v2.x message queuing, allowing records from different EHR vendors to share a single normalized representation.
- **End-to-End Encryption with RBAC** — AES-256 at rest, TLS 1.3 in transit, four enforced roles per API request, and 90-day key rotation through cloud KMS.

What makes PatientPath significant is the combination of three properties that competing solutions do not offer together. First, the platform is EHR-agnostic and bridges existing systems rather than replacing them, meaning organizations adopt PatientPath without a multi-million-dollar EHR migration. Second, offline sync resilience makes the platform usable in rural and low-bandwidth clinics, which traditional digital fax replacements and cloud-only HIE solutions cannot serve. Third, the single-form referral workflow with immediate consent-to-access tokenization collapses what is currently a multi-step manual process into a single transaction, directly eliminating the bottlenecks responsible for the failure of fax-based transfers in the first place.

2.2 Major Components (Hardware/Software)

PatientPath does not require dedicated hardware on the customer side. The platform is hosted on commodity cloud infrastructure (AWS, Azure, or GCP, selected per deployment), and

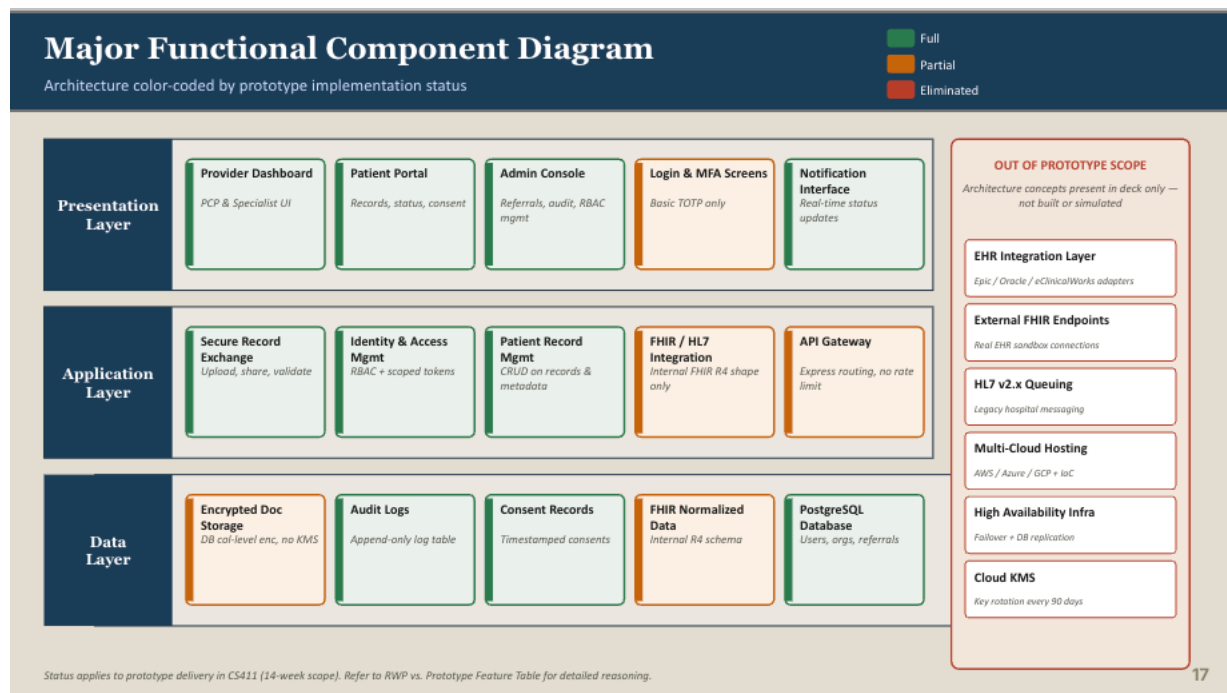
end users access the system through standard web browsers on any current desktop, tablet, or mobile device. Production deployment requires a load-balanced application tier behind a cloud-managed API gateway, an active-passive PostgreSQL cluster with replication across availability zones for high availability, encrypted object storage for document blobs, and a managed key management service (AWS KMS or Azure Key Vault) for AES-256 key custody and rotation. Cached reads and queued writes for offline mode are held client-side in the browser through Service Worker storage and IndexedDB, requiring no installed software.

The software is structured according to the three-layer Major Functional Component Diagram (MFCD) defined in the team's design deck. The Presentation Layer contains the four user-facing interfaces — Provider Dashboard, Patient Portal, Admin Console, and Login/MFA screens — along with the notification interface, all rendered as a React single-page application. The Application Layer sits behind a single API gateway and contains the platform's business logic, organized into four services: the Secure Record Exchange Service, Identity and Access Management, Patient Record Management, and the FHIR/HL7 Integration Service. The Data Layer holds the encrypted document storage for medical records, a PostgreSQL database for users and organizations, a separate FHIR-normalized data store, an append-only audit log database, and the consent records store.

The software stack consists of TypeScript with React on the front end and TypeScript with Node.js and Express on the back end. PostgreSQL serves as the primary relational database, accessed through Prisma as the ORM. WebSockets push real-time status updates to dashboards. Authentication is built on Speakeasy for TOTP-based MFA and OAuth 2.0 / OpenID Connect for hospital SSO integration. Encryption uses TLS 1.3 in transit and AES-256 at rest, with keys held in AWS KMS or Azure Key Vault. CI/CD runs through GitHub Actions, with Jest as the

testing framework, and source control is managed in GitHub. The interoperability layer is built against the HL7 FHIR R4 specification with HL7 v2.x message support for legacy EHR queues.

Figure 3
Major Functional Component Diagram



3 Identification of Case Study

PatientPath is being developed for the small-to-mid-size primary care practices, specialty clinics, and federally qualified health centers (FQHCs) that experience the highest cost from fragmented record exchange and the lowest level of meaningful interoperability. As established in Section 1, only 16% of hospitals routinely share records with long-term and post-acute providers outside their own networks, and 71% of office-based physicians work with referral partners who lack any cross-organization exchange capability (ASTP/ONC, 2024; ONC, 2022). These figures concentrate in rural and lower-resourced settings where IT infrastructure is constrained and connectivity is unreliable — the exact populations for whom every additional

fax round-trip translates into delayed care. PatientPath targets this segment because it is the segment that benefits most from a bridge-style interoperability platform that does not require an EHR replacement and remains usable when network connectivity is intermittent.

The case study group for prototype evaluation will consist of a representative cross-section of the four primary user roles: one to two primary care providers initiating referrals, one to two specialists receiving them, one administrative staff member responsible for referral coordination and record validation, one system administrator with responsibility for user provisioning and audit review, and two to three patient participants exercising the consent and transfer-visibility workflows from the patient portal. This group will be drawn from a single rural community health network so that the prototype can be evaluated end-to-end on a complete referral loop — from the patient encounter through consent, transmission, receipt, and acknowledgment — within one organizational boundary the team can directly observe and instrument.

Once validated against this initial group, PatientPath is positioned for expansion to a broader user base that shares the same structural pain points. Future users include behavioral health providers (only 17% of which routinely share records today), long-term and post-acute care facilities, multi-specialty group practices, regional Health Information Exchanges (HIEs) seeking a low-friction onboarding option for smaller member organizations, and larger hospital systems that need to coordinate referrals with the rural and specialty practices in their catchment area. The plugin-based EHR integration model is intended to support this expansion without requiring core platform changes for each new participating organization.

4 Prototype Description

The PatientPath prototype is a scoped implementation built to demonstrate the platform's core value proposition — secure, real-time record sharing and closed-loop referral coordination — within a controlled environment that the case study group can exercise end-to-end. The prototype implements the same three-layer architecture as the end product (Presentation, Application, Data) and uses the same software stack (TypeScript, React, Node.js, PostgreSQL, Prisma) so that lessons from prototype evaluation transfer directly to the production build. What the prototype omits is the breadth, redundancy, and external integration surface that production deployment requires; it does not omit the security, consent, or audit functions, which are present in the prototype because removing them would invalidate the workflows being evaluated.

The prototype implements the seven user-facing workflows that anchor the design: secure record upload with AES-256 encryption, single-form referral creation with automatic specialist notification, real-time referral status tracking with stale-referral alerts, patient consent grant and revocation, multi-factor authentication, offline cache and sync queue for record reads, and the immutable audit log viewable from the admin console. All four user roles — primary care provider, specialist, administrative staff, and patient — are functional, with role-based access control enforced at the API gateway. Records are exchanged in FHIR R4 normalized form within a sandbox EHR data set that simulates the variety of source-system formats the production platform must handle.

The prototype differs from the end product along five specific dimensions. First, EHR integration uses simulated EHR data rather than live adapters to Epic, Oracle, or eClinicalWorks; the production EHR adapter layer requires per-vendor authentication, sandbox certification, and

customer-specific configuration that is out of scope for prototype evaluation. Second, the prototype runs in a single-region cloud deployment without active-passive failover or multi-AZ database replication; production operates against a 99.9% uptime SLA that requires the full HA configuration. Third, external services for key management, identity provider integration, and notification delivery use sandbox or development-tier services rather than production-grade KMS, hospital SSO, and SMS/email gateways. Fourth, the prototype is provisioned only for the case study group, while production supports unbounded onboarding through self-service organization registration. Fifth, the prototype demonstrates HIPAA-aligned controls but has not undergone formal third-party HIPAA audit, SOC 2 Type II certification, or external penetration testing, all of which are required before any handling of real PHI in production. Table 1 summarizes these differences.

Table 1

Comparison Between Real-World Product (RWP) and Prototype

Dimension	Real-World Product	Prototype
EHR Integration	Live adapters for Epic, Oracle, and eClinicalWorks via FHIR R4 / HL7 v2.x	Simulated EHR data set in FHIR R4 format; no live vendor APIs
Deployment	Multi-region, multi-AZ, active-passive failover; 99.9% uptime SLA	Single-region cloud deployment; no failover
Key Management	Production AWS KMS / Azure Key Vault with 90-day rotation	Sandbox KMS instance; manual rotation
Identity Provider	Hospital SSO via OAuth 2.0 / OpenID Connect	Local credential store with TOTP MFA
User Population	Unbounded; self-service organization onboarding	Case study group only (~7 users)

Notifications	Production SMS and email gateways	Logged in-app notifications only
Compliance	HIPAA audit, SOC 2 Type II, annual penetration testing	HIPAA-aligned controls implemented; no formal certification
Scale	Horizontal autoscaling under load	Fixed development-tier capacity
Offline Sync	Full reconciliation with server-side conflict resolution at scale	Demonstrable cache + queue against single test database
Audit Log	Append-only, exportable as CSV/PDF, long-term retention	Append-only, viewable in admin console; short retention

5 Glossary

- **EHR:** Electronic Health Record
- **HIPPA:** Health Insurance Portability and Accountability Act
- **FHIR:** Fast Healthcare Interoperability Resources (HL7 standard for health data exchange)
- **HL7:** Health Level Seven (International healthcare data standards organization)
- **RBAC:** Role-Based Access Control
- **MFA:** Multi-Factor Authentication
- **API:** Application Programming Interface
- **REST:** Representational State Transfer (architectural style for APIs)
- **HIE:** Health Information Exchange
- **PCP:** Primary Care Provider
- **FQHC:** Federally Qualified Health Center
- **CI/CD:** Continuous Integration / Continuous Deployment

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